

RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS

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THE LIKELIHOOD OF DIFFERENT OUTCOMES

In Articles No.8-10, we looked into the basics needed to calculate probabilities, including combinatorics and conditional probability. While these techniques allow us to determine the likelihood of events, we are often interested in something more specific; the value of an uncertain quantity. For example, we may wish to model the number of defaults in a loan portfolio, the return of an asset over the next day, or the value of an option at expiry. To study such, we can introduce the concept of a random variable and its associated probability distribution.

Probability distributions provide a mathematical description of how likely different outcomes are to occur. Broadly speaking, distributions can be classified as either discrete or continuous, depending on the nature of the possible outcomes. Understanding this distinction is fundamental to probability theory and forms the basis of many models used in statistics, machine learning, *etc.*

RANDOM VARIABLES

Random variables can be classified into two categories, discrete and continuous. A discrete random variable can take a finite or countably infinite set of values, such as the number of heads obtained in a series of coin tosses. In contrast, a continuous random variable can take any value within a given interval, such as the future price of a stock or the time until a particular event occurs. This distinction is important, as the methods used to describe and analyse these variables differ significantly.

To describe discrete and continuous random variables, we use the following definitions and properties:

Random Variable (X)	Discrete	Continuous
Cumulative Distribution Function (CDF)	$F(a) = P(X \leq a)$	$F(a) = \int_{-\infty}^a f(x)dx$
Probability Mass Function (PMF) Probability Density Function (PDF)	PMF: $p(x) = P(X = x)$	PDF: $f(x) = \frac{d}{dx}F(x)$
Expected Value ($\mathbb{E}[X]$)	$\sum_{x:p(x)>0} xp(x)$	$\int_{-\infty}^{\infty} xf(x)dx$
Expected Value of $g(X)$ ($\mathbb{E}[g(X)]$)	$\sum_{x:p(x)>0} g(x)p(x)$	$\int_{-\infty}^{\infty} g(x)f(x)dx$
Variance ($Var[X]$)	$\mathbb{E}[(X - \mathbb{E}[X])^2]$	
Standard Deviation (σ)	$\sqrt{Var[X]}$	

Note that for continuous random variables, $P(X = x) = 0$, $\forall x \in (-\infty, \infty)$, thus $P(X \leq x) = P(X < x)$.

DISCRETE PROBABILITY DISTRIBUTIONS

There are numerous probability distributions, however, some are more widely-used than others. Here are a few of the following:

Name	PMF	$\mathbb{E}[X]$	$Var[X]$
Uniform	$P(x) = \frac{1}{b-a+1}$, $x = a, a+1, \dots, b$	$\frac{b+a}{2}$	$\frac{(b-a+1)^2-1}{12}$
Binomial	$P(x) = \binom{n}{x}p^x(1-p)^{n-x}$, $x = 0, 1, \dots, n$	np	$np(1-p)$
Poisson	$P(x) = \frac{e^{-\lambda t}(\lambda t)^x}{x!}$, $x = 0, 1, \dots$	λt	λt
Geometric	$P(x) = (1-p)^{x-1}p$, $x = 1, 2, \dots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative Binomial	$P(x) = \binom{x-1}{r-1}p^r(1-p)^{n-r}$, $x = r, r+1, \dots$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$

The discrete uniform R.V. represents the occurrence of a value between a number a and b when all values in the set $\{a, a+1, \dots, b\}$ have equal probability. The Binomial R.V. represents the number of successes in a sequence of n experiments when each trial is independently a success with probability p . The Poisson R.V. represents the number of events occurring in a fixed period of time with the expected number of occurrences, λt , when events occur with a known average rate, λ , and are independent of time since the last event. Geometric R.V. represents the trial number, n , to get the first success when each trial is independently a success with probability, p . The negative Binomial R.V. represents the trial number to get to the r^{th} success when each trial

is independently a success with probability, p .

CONTINUOUS PROBABILITY DISTRIBUTIONS

Name	PDF	$\mathbb{E}[X]$	$Var[X]$
Uniform	$\frac{1}{b-a}, a \leq x \leq b$	$\frac{b+a}{2}$	$\frac{(b-a)^2}{12}$
Normal	$\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, x \in (-\infty, \infty)$	μ	σ^2
Exponential	$\lambda e^{-\lambda x}, x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gamma	$\frac{\lambda e^{-\lambda x} (\lambda x)^{\alpha-1}}{\Gamma(\alpha)}, x \geq 0, \Gamma(a) = \int_0^\infty e^{-y} y^{a-1} dy$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
Beta	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, 0 < x < 1$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2}$

The continuous uniform R.V. describes a R.V. uniformly distributed over the interval $[a, b]$. Because of the central limit theorem, the normal distribution, also known as the Gaussian distribution is the most famous continuous distribution. The exponential distribution models the arrival time of an event if it has a constant arrival rate, λ . The Gamma distribution with parameters (α, λ) is normally described as the distribution of the amount of time one has to wait until a total of n events to occur. Beta distributions are used to model events that are constrained within a defined interval. By adjusting the shape of parameters, α and β , it can model different shapes of probability density functions.

AN EXAMPLE

There are many examples to choose from, considering the breadth of distributions to choose from, but one example I particularly like is the meeting problem. Two bankers each arrive at the station at some random time between 5 and 6AM, where the arrival time for either is uniformly distributed. They stay exactly 5 minutes and then leave. What is the probability they will meet on a given day?

Solution: Assume banker A arrives X minutes after 5AM and B arrives Y minutes after 5AM. X and Y are independent uniform distribution between 0 and 60. Since both only stay exactly 5 minutes, A and B meet if and only if $|X - Y| \leq 5$. Thus, the probability that A and B will meet is simply the area where $|X - Y| \leq 5$ divided by a 60 by 60 square. That is, $P = \frac{60 \times 60 - 2 \times (\frac{1}{2} \times 55 \times 55)}{60 \times 60} = \frac{23}{144}$.

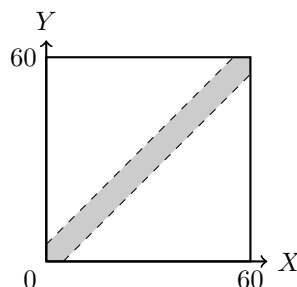


Figure 1: Distribution of A 's and B 's Arrival Times

WHY DISTRIBUTIONS MATTER IN FINANCE

Probability distributions are fundamental to quantitative finance because they describe the behaviour of uncertain market variables. For example, in the articles on the Black-Scholes series, asset prices are assumed to follow a geometric Brownian motion, GBM, implying that future asset prices are lognormally distributed. The distribution of future prices allows us to calculate probabilities, expected payoffs, and ultimately the fair value of financial derivatives.

More generally, distributions provide the mathematical foundation for modelling asset returns, measuring risk, and pricing securities. By understanding the distribution of a random variable, we gain insight not only into its expected value but also the range and likelihood of possible outcomes.

CONCLUSION

Probability distributions provide a mathematical framework for describing uncertainty. Whether discrete or continuous, they form the foundation for modelling, analysing, and making decisions under uncertainty in quantitative finance.